# Multitask Sparse Representation Model-Inspired Network for Hyperspectral Image Denoising

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***Abstract*— Hyperspectral images (HSIs) are prone to noise because of the imaging mechanism and environment. This article proposes a multitask sparse representation (MTSR) model-inspired neural network for HSI denoising. Unlike other deep learning (DL)-based methods, our network is interpretable, whose network architecture is induced by unfolding the iterative optimization of an MTSR model. On one hand, the model globally represents the common structure among bands, such as image edges, with the shared sparse coefficients. On the other hand, it separately encodes the unique structure of individual bands with unshared ones to capture image details. Accordingly, our network has three modules: the shared sparse representation (SSR) module, the unshared sparse representation (USR), and the image reconstruction (IR) module. All the modules are connected with a specific operation of the iterative optimization algorithm, equipping the network with clear physical interpretation. Experimental results on both the synthetic and real-world datasets demonstrate the superior performance of our method, visually and quantitatively. The codes will be publicly available at https://github.com/bearshng/mtsrnn for reproducible research.**

***Index Terms*— Deep unfolding, hyperspectral image (HSI) denoising, multitask learning, sparse representation (SR).**

## NOMENCLATURE

| | |
|---|---|
| $\mathbf{Y}$ | 2-D matrix of the noisy HSI of size $N \times B$. |
| $\mathbf{X}$ | 2-D matrix of the clean HSI of size $N \times B$. |
| $\mathbf{G}$ | 2-D matrix of the noise term of size $N \times B$. |
| $N$ | Number of pixels. |
| $B$ | Number of bands. |
| $p$ | Dimensionality of each atom. |
| $m$ | Number of atoms in the dictionary for shared structure. |
| $q$ | Number of atoms in the dictionary for unshared structure. |
| $k$ | Patch index. |
| $\mathbf{R}_k$ | Operator that extracts the $k$th patch from a given image. |
| $\mathbf{R}_k^{\mathrm{T}}$ | Operator that puts the $k$th patch back to the original position. |
| $\mathbf{A}_b$ | $p \times m$ dictionary at the $b$th band for shared sparse representation. |
| $\mathbf{D}_b$ | $p \times q$ dictionary at the $b$th band for unshared sparse representation. |
| $\mathbf{s}_k$ | Shared sparse representation of the $k$th patch. |
| $\mathbf{u}_{bk}$ | Unshared sparse representation of the $k$th patch of the $b$th band. |
| $\lambda_k$ | Parameter that controls the degree of sparsity of shared structure. |
| $\mu_k$ | Parameter that controls the degree of sparsity of unshared structure. |
| $L$ | Step size. |
| $\|\cdot\|_F$ | Frobenius norm. |
| $\|\cdot\|_1$ | $L_1$ norm. |
| $\|\cdot\|_0$ | $L_0$ norm. |
| $\eta(\cdot)$ | Soft-thresholding operation. |
| $\mathbf{W}_{bs}$ | $m \times p$ learnable parameter related to $\mathbf{A}_b^T$. |
| $\mathbf{V}_{bs}$ | $p \times m$ learnable parameters related to $\mathbf{A}_b$. |
| $\mathbf{E}_{bs}$ | $p \times m$ learnable parameters related to $\mathbf{A}_b$ |
| $\mathbf{W}_{bu}$ | $q \times p$ learnable parameters related to $\mathbf{D}_b^T$. |
| $\mathbf{V}_{bu}$ | $p \times q$ learnable parameters related to $\mathbf{D}_b$. |
| $\mathbf{E}_{bu}$ | $p \times q$ learnable parameters related to $\mathbf{D}_b$ |
| $\boldsymbol{\mu}_t$ | Learnable thresholding parameters at the $t$th layer, related to $\mu_k$. |
| $\lambda_t$ | Learnable thresholding parameters at the $t$-th layer, related to $\lambda_k$. |
| $\Theta$ | Parameter set of MTSNN++. |
| $T$ | Number of unfolding layers. |
| $\mathcal{L}(\cdot)$ | Loss function of MTSNN++. |

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## I. INTRODUCTION

HYPERSPECTRAL images (HSIs) can reveal more physical characteristics through the per-pixel irradiance measurements in multiple wavelengths. The spectral information makes HSIs have advantages over commonly used color images such as material identification [1], remote sensing [2], [3], object detection [4], and tracking [5], [6]. Because of shorter exposure time, imaging environment, and sensor limitation, the acquired HSIs are often degraded by noises, dra-

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matically deteriorating the utility of HSIs in downstream tasks. Moreover, because of the differences in sensor sensitivity, the noise identities are different among bands. The nonindependent and identically distributed (non-i.i.d.) noise makes it more challenging than color images. Therefore, how to recover the clean HSI from noisy observation is an important problem and indispensable for the deployment of hyperspectral imaging systems in many real-world applications.

The existing HSI denoising methods can be categorized into model-based and learning-based. The model-based methods establish the relationship between the underlying clean HSIs and noisy HSIs and carry out denoising based on the physical priors related to image formation or natural statistics [7], [8], [9], [10]. The model-based methods have high interpretability and theoretical guarantee but are limited by the hand-crafted prior. The learning-based methods, in particular deep learning (DL), remove noises without an explicit model and learn the nonlinear mapping between noisy–clean HSI pairs [11]. Provided with large-scale data, the learning-based methods often outperform conventional model-based approaches in terms of performance and efficiency. One of the prominent drawbacks of the learning-based methods is their "black-box" property, making it difficult to understand their internal denoising mechanism. Therefore, it is preferable to develop a "glass-box" network that combines the strengths of the model-based and learning-based methods for interpretable denoising.

Based on the deep unfolding technique, model-inspired networks can fulfill the aforementioned goal and have recently gained extensive attention in image processing. The basic idea is to cast the update rules of the iterative optimization in the model-based methods as a stack of recurrent blocks and transform the optimization into a deep neural network by introducing some learnable parameters including hyperparameters. In this way, instead of iterative update and hyperparameter tuning, the solution of the model-based methods is learned with deep neural networks. By learning from the data, the trained network may outperform the original algorithm [12]. Thanks to the hybrid advantages of the model-based and learning-based methods, the model-inspired networks are equipped with interpretability and better generalization ability. As such, plenty of model-inspired networks have been applied in hyperspectral unmixing [13], fusion [14], [15], and anomaly detection [16] and achieved impressive performance in both efficiency and effectiveness.

Following the above line, this article studies HSI denoising from the physical modeling to the development of learnable network architects. From the model aspect, regarding denoising in an individual band as a single task, denoising for all the bands is a multitask problem. All the tasks are globally linked by the shared structures but differentiated by unshared structures to simultaneously capture the shared global spectral correlation among bands and band-specific properties. Accordingly, we build a multitask sparse representation (MTSR) model to depict the common structures with shared sparse coefficients and unique structures with unshared ones. In this way, the spatial and spectral priors of HSIs are effectively exploited. From the network aspect, the optimization algorithm of the MTSR model is further unfolded to yield an MTSR model-inspired neural network named MTSNN++, by introducing the dictionaries and the hyperparameters as learnable parameters. As shown in Fig. 1, all the network modules are closely linked with the MTSR model, conveying certain interpretability. Such interpretability makes the prior knowledge of the model be easily integrated into the network to enhance the representation ability of the network. We test MTSNN++ on both the synthetic data and real-world data against both the model-based and DL-based methods. Extensive experiments show that our MTSNN++ achieves state-of-the-art denoising.

A preliminary conference version of this work was presented in ICASSP 2022 [17]. In the conference version, all the bands are linked by the shared sparse coefficients to depict the common structures among bands. In contrast, this article introduces a more robust physical model which depicts not only the common structures but also the unique structures of individual bands, facilitating recovering image details. Moreover, more comprehensive comparisons of different datasets, analysis of network parameters, and visual results are provided to illustrate the advantages of our work.

The following of this article is organized as follows. Section II introduces the related works. The details of the proposed methods are given in Section III. Section IV presents the experimental results. This article concludes in Section V.

## II. Related Work

HSI denoising is a typical ill-posed inverse problem. Model-based and DL-based methods are two main classes. This section introduces the recent advances of two classes of methods.

### A. Model-Based Methods for HSI Denoising

The model-based methods take advantage of different hand-crafted priors of HSIs to tackle HSI denoising. HSIs can be regarded as a stack of gray images taken at different wavelengths. Gray image denoising methods can be applied to HSIs in a band-by-band manner. HSIs are 3-D data with the spatial and spectral dimensions strongly correlated and noises exist in both the spatial and spectral domains of HSIs. Compared with band-by-band denoising, the spatial–spectral methods which jointly exploit both spatial and spectral information dramatically improve the denoising performance [18], [19]. For example, to exploit the spatial–spectral dependency among HSIs, total variation is extended to the spatial–spectral one that punishes both the spatial and spectral differences [20], [21], [22], producing higher denoising performance.

HSIs feature global spectral redundancy, and local and nonlocal spatial redundancy [23], [24]. This powerful redundancy prior can be combined with the spatial–spectral correlation prior of HSIs to encourage more robust HSI denoising. Low-rank representation is a preferable manner to encode the spatial–spectral redundancy of HSIs [23], [25], [26], [27], [28], [29], [30]. For example, He et al. [31] used matrix rank minimization to encode the spatial low-rankness. However, the matrix-based methods require converting 3-D HSIs into

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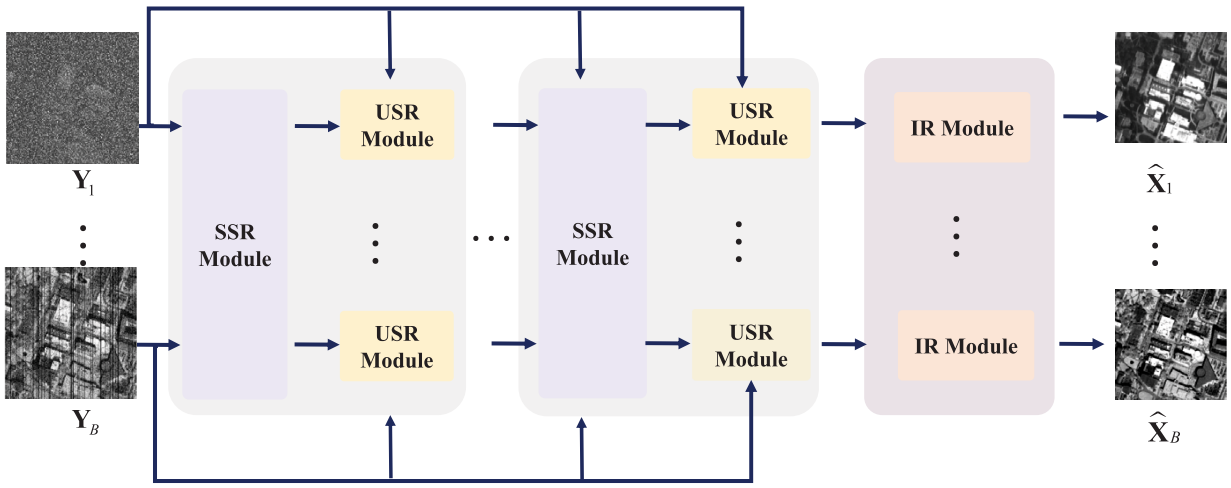



Fig. 1. Framework of the proposed MTSNN++ method. The SSR and USR modules extract the shared common sparse features and unshared unique sparse features, respectively. The IR module reconstructs the clean HSI from extracted features.

2-D data, leading to spatial information loss. By modeling HSIs as 3-D tensors, the tensor-based methods can better capture the spatial–spectral property of HSIs and achieve more promising denoising [32], [33], [34]. For example, Zhang et al. [35] performed the tensor ring decomposition in the transformed space to characterize the global low-rankness, boosting the denoising performance. Sun et al. [9] adopted a three-way log-based tensor nuclear norm to exploit the global spectral–spatial low-rankness and group sparsity in subspace to explore the nonlocal self-similarity. Peng et al. [10] performed low-rank matrix factorization in the gradient domain to simultaneously exploit the global spectral low-rankness and local spatial–spectral correlation. Unfortunately, the difficulties of rank determination threaten their practical usage for real-world HSI denoising.

Besides low-rank representation, spatial–spectral sparse coding is another way to depict the redundancy of HSIs [36]. By representing each 3-D hyperspectral cube with the weighted sum of a few atoms in a dictionary, the spatial–spectral methods can better depict the local redundancy. When considering the nonlocal redundancy, the denoising performance can be enhanced. To this end, structural spatial–spectral sparse representation (SR) [28], [37], [38] groups similar 3-D cubes into clusters and adopted different norms such as the $L_{1,2}$ norm [39] on the sparse coding matrix to jointly recover different cubes within clusters.

Modeling HSIs with 3-D local cubes breaks the spectral continuity and prevents the global spectral correlation from being effectively captured. Because all the bands of HSIs are captured in the same scene, they tend to share some common structures, for example, image edges. To this end, Ye et al. [40] modeled each band with 2-D patches and enforced all the patches in the same spatial location to have the same sparsity level to capture the global structure shared among bands. However, this method ignores that a single band also has unique structures related to image details due to sensor sensitivity at different wavelengths, resulting in inferior performance. To this end, this article adopts an MTSR model which represents HSIs with the common sparse component among all the bands and the specific sparse component in a single band.

### B. DL-Based Methods for HSI Denoising

DL tackles HSI denoising with a data-driven strategy and has become a hotspot in image denoising, thanks to its competent representation ability. With enough data, the DL-based methods can learn more robust implicit image priors and overcome the limitation of the model-based methods, such as time-consuming optimization and hyperparameter tuning. Currently, the main efforts are paid to design various network architectures to learn a rich feature representation of HSIs for noise removal. 2-D convolutions [41], [42] can only extract spatial correlations and cannot exploit spatial–spectral correlations, yielding poor denoising performance. Instead of 2-D convolutions, 3-D convolution performs computation in both the spatial and spectral domains to depict the spatial–spectral correlation among HSIs [43], [44], maintaining more spatial–spectral details.

Besides local information, nonlocal contextual information is also essential to extract robust features and can be captured by the attention mechanism [45], [46], [47]. By modeling the similarity among different spatial locations, the resulting HSI has clearer contents. The denoising performance can be strengthened by the consideration of the global spectral correlation. The wavelengths are continuous and the HSIs can be regarded as sequential data. Therefore, [24] and [48] took advantage of recurrent neural networks to successively strengthen the correlation between adjacent spectra. Alternatively, different spectral bands can be viewed as vertices on the graph. For this regard, Cao et al. [49] modeled the global spectral correlation with graph neural networks. Moreover, Chen et al. [50] introduced a transformer-based network which uses the multihead global spectral attention to model global correlations along the spectrum. Li et al. [11] developed rectangle self-attention horizontally and vertically to capture nonlocal similarities in the spatial domain and spectral enhancement to extract global low-rank properties of the

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spatial–spectral cube to suppress noise. Thanks to the strong ability in long-range relationship modeling, transformers usually produce higher denoising effectiveness but at a cost of more parameters.

The network architecture of most DL-based methods is built heuristically, requires troublesome trials, and lacks interpretability. It is difficult for the DL-based methods to share the achievements of the model-based methods. In our work, we transform the optimization of our MTSR model into a model-inspired network architecture. Accordingly, the merits of the model-based methods and DL-based methods can be effectively inherited while suppressing their disadvantages.

## III. MTSR Model-Inspired Network

We present the details of our method in this section including the mathematical model, optimization procedure, unfolded network, and implementation issues. Before introducing our model and network, the related major notations are listed in Nomenclature.

### A. Multitask Sparse Representation Model

Let $\mathbf{X}$ be a clean HSI with $N$ pixels and $B$ bands. When degraded with the additive noises $\mathbf{G} \in \mathbb{R}^{N\times B}$, the measured noisy HSI $\mathbf{Y} \in \mathbb{R}^{N\times B}$ can be described as follows:

$$\mathbf{Y} = \mathbf{X} + \mathbf{G}. \tag{1}$$

It is known that extracting $\mathbf{X}$ from $\mathbf{Y}$ is very challenging, and the solution space is huge because of its ill-posed nature. The prior knowledge which describes how $\mathbf{X}$ is formed is a beneficial clue to make the inverse problem well-posed. Based on this observation, we derive a physical model embodying the physical property of $\mathbf{X}$. The strong spatial local correlation implies spatial redundancy. Sparsity is a widely accepted tool to depict signal redundancy, leading to the SR model [51], [52]. The SR model assumes that each local patch of $\mathbf{X}_b$ is formulated as a combination of few atoms in a prespecified dictionary $\mathbf{A}_b \in \mathbb{R}^{p\times m}$ where $p$ and $m$ represent the dimensionality of each atom and the total number of atoms, respectively [53].

Let $\mathbf{R}_k$ denote an operator extracting the $k$th patch from $\mathbf{Y}_b$. Under the framework of the SR model, the denoising problem of the $b$th band can be depicted by

$$\min_{\{\mathbf{s}_{bk}\}} \sum_k \left( \frac{1}{2}\|\mathbf{R}_k\mathbf{Y}_b - \mathbf{A}_b\mathbf{s}_{bk}\|_F^2 + \mu_k\|\mathbf{s}_{bk}\|_0 \right) \tag{2}$$

where $\mathbf{s}_{bk} \in \mathbb{R}^m$ is a sparse coefficient vector, and $\{\mu_k\}$ are hyperparameters determining the impact of the sparsity constraint.

Since all the bands describe the context of the same scene, they are expected to share some common structures, for example, image edges. In addition, even captured in the same scene, due to the sensor sensitivity at different wavelengths, the spatial response may be different. In other words, there are also some unique structures across bands, for example, fine image details. To this end, we enforce that all the bands have the same sparse coefficients at the same spatial position to depict the common structures globally and also have their unique sparse coefficients to allow for the unshared unique structures. In this way, the denoising problem for all the bands can be formulated as follows:

$$\min_{\{\mathbf{s}_k,\mathbf{u}_{bk}\}} \sum_b \sum_k \left( \frac{1}{2}\|\mathbf{R}_k\mathbf{Y}_b - \mathbf{A}_b\mathbf{s}_k - \mathbf{D}_b\mathbf{u}_{bk}\|_F^2 \right) + \sum_b \sum_k \lambda_k\|\mathbf{u}_{bk}\|_0 + \sum_k \mu_k\|\mathbf{s}_k\|_0 \tag{3}$$

where $\mathbf{s}_k \in \mathbb{R}^m$ is the coefficient shared across all the bands. $\mathbf{D}_b \in \mathbb{R}^{p\times q}$ and $\mathbf{u}_{bk} \in \mathbb{R}^q$ are the unshared dictionary and sparse coefficient, respectively. $\mu_k$ and $\lambda_k$ control the sparse level of shared and unshared structures, respectively. For simplicity, we use the same parameters $\lambda_k$ for all the bands. In this way, we obtain an MTSR model. The noise removal of an individual band is a single task, and the noise removal of all the bands is an MTSR problem.

### B. Optimization Procedure

The NP-hardness of the regularized $L_0$ minimization problem makes (3) intractable. Alternatively, (3) can be relaxed to the following $L_1$ norm penalized minimization problem:

$$\min_{\{\mathbf{u}_{bk},\mathbf{s}_k\}} \sum_b \sum_k \left( \frac{1}{2}\|\mathbf{R}_k\mathbf{Y}_b - \mathbf{A}_b\mathbf{s}_k - \mathbf{D}_b\mathbf{u}_{bk}\|_F^2 \right) + \sum_b \sum_k \lambda_k\|\mathbf{u}_{bk}\|_1 + \sum_k \mu_k\|\mathbf{s}_k\|_1 \tag{4}$$

where $\|\cdot\|_1$ is the convex $L_1$ norm encouraging sparsity.

Assuming the underlying dictionaries are known at the moment, the optimization problem can be further decomposed into the following two subproblems corresponding to calculating $\mathbf{u}_{bk}$ and $\mathbf{s}_k$:

$$\begin{aligned} &\arg\min_{\mathbf{s}_k} \sum_b \sum_k \left( \frac{1}{2}\|\mathbf{R}_k\mathbf{Y}_b - \mathbf{A}_b\mathbf{s}_k - \mathbf{D}_b\mathbf{u}_{bk}\|_F^2 \right) + \sum_k \mu_k\|\mathbf{s}_k\|_1 \\ &\arg\min_{\mathbf{u}_{bk}} \sum_b \sum_k \left( \frac{1}{2}\|\mathbf{R}_k\mathbf{Y}_b - \mathbf{A}_b\mathbf{s}_k - \mathbf{D}_b\mathbf{u}_{bk}\|_F^2 + \lambda_k\|\mathbf{u}_{bk}\|_1 \right). \end{aligned} \tag{5}$$

Both the problems are the standard $L_1$ norm regularized SR problems. It finds the optimal sparse code vector that simultaneously minimizes the reconstruction error and the $L_1$ sparsity penalty on the code

$$\min_{\alpha} \frac{1}{2}\|\mathbf{z} - \mathbf{D}\alpha\|_F^2 + \lambda\|\alpha\|_1 \tag{6}$$

where $\mathbf{z}$ is the given input vector, $\mathbf{D}$ is the dictionary, and $\alpha$ is the sparse vector. This problem can be solved by the well-known iterative shrinkage and thresholding algorithm (ISTA) [54] with the following iterative form:

$$\alpha^{(t+1)} = \text{soft}_{\lambda/L}\left( \alpha^{(t)} + \frac{1}{L}\left(\mathbf{D}^{\mathrm{T}}\left(\mathbf{z} - \mathbf{D}\alpha^{(t)}\right)\right) \right) \tag{7}$$

where $L$ is the step size; $\text{soft}_\theta(x)$ is the soft-thresholding operator defined as $\text{soft}_\theta(x) = \text{sign}(x)\max(|x| - \theta, 0)$; and $t$ indexes the iteration number. Following the update rule in (7), we now alternately update $\mathbf{s}_k$ and $\mathbf{u}_{bk}$ with the other variables fixed.

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*1) Update* $\mathbf{s}_k$*:* Let $\mathbf{P}_b = \mathbf{R}_k\mathbf{Y}_b - \mathbf{D}_b\mathbf{u}_{bk}$. The optimization problem of $\mathbf{s}_k$ can be expressed as

$$\min_{\mathbf{s}_k} \sum_k \sum_b \left( \frac{1}{2}\|\mathbf{P}_b - \mathbf{A}_b\mathbf{s}_k\|_F^2 + \mu_k\|\mathbf{s}_k\|_1 \right). \tag{8}$$

The update rule amounts to

$$\mathbf{s}_k^{(t+1)} = \text{soft}_{\mu_k/L_s}\left( \mathbf{s}_k^{(t)} + \sum_b \frac{1}{L_s}\mathbf{A}_b^{\mathrm{T}}\left(\mathbf{P}_b - \mathbf{A}_b\mathbf{s}_k^{(t)}\right)\right) \tag{9}$$

where $L_s$ is the step size.

*2) Update* $\mathbf{u}_{bk}$*:* By defining $\mathbf{Q}_b = \mathbf{R}_k\mathbf{Y}_b - \mathbf{A}_b\mathbf{s}_k$, the optimization problem can be decoupled into $B$ similar problems with each one being

$$\sum_k \left( \frac{1}{2}\|\mathbf{Q}_b - \mathbf{D}_b\mathbf{u}_{bk}\|_F^2 + \lambda_k\|\mathbf{u}_{bk}\|_1 \right). \tag{10}$$

This problem has the following solution:

$$\mathbf{u}_{bk}^{(t+1)} = \text{soft}_{\lambda_k/L_{bu}}\left( \mathbf{u}_{bk}^{(t)} + \frac{1}{L_{bu}}\left(\mathbf{D}_b^{\mathrm{T}}\left(\mathbf{Q}_b - \mathbf{D}_b\mathbf{u}_{bk}^{(t)}\right)\right)\right) \tag{11}$$

where $L_{bu}$ is also the step size.

*3) Image Reconstruction:* Running (9) and (11) for numerous times yields the solutions of $\mathbf{s}_k$ and $\mathbf{u}_{bk}$. The clean HSIs can then be obtained by averaging the cleaned version of overlapping patches at each position. Mathematically, the reconstructed HSI can be generated by

$$\widehat{\mathbf{X}}_b = \left(\sum_k \mathbf{R}_k^{\mathrm{T}}\mathbf{R}_k\right)^{-1} \sum_k \mathbf{R}_k^{\mathrm{T}}(\mathbf{A}_b\mathbf{s}_k + \mathbf{D}_b\mathbf{u}_{bk}). \tag{12}$$

In general, the update rules in (9), (11), and (12) are computationally inefficient because of the time-consuming iterative optimization. Regarding the HSI denoising task, the hyperparameters $\{\mu, \lambda\}$ selection is also troublesome. In Section III-C, we will derive a neural network based on the three update rules to solve the above issues.

### C. Network Architecture

As shown in Fig. 1, our MTSNN++ is made up of three fundamental modules, i.e., the shared sparse representation (SSR) module, unshared sparse representation (USR) module, and image reconstruction (IR) module. The SSR and USR modules fulfill the MTSR to extract the shared and unique sparse structures from the given noisy HSIs. The IR module is responsible for recovering the clean HSIs from the extracted sparse structures. All the components are seamlessly connected with the update rules in (9), (11), and (12).

As mentioned above, (9) and (11) share a very similar form with (7). Iteratively running (7) can be seen as a feedforward process of a neural network with $T$ layers. By untying the variables and introducing learnable parameters $\mathbf{W}$, $\mathbf{E}$, and $\lambda$, the equation can be unfolded into a network whose typical layer is formally expressed as

$$\alpha^{(t+1)} = \eta_{\lambda_t}(\alpha^{(t)} + \mathbf{W}(\mathbf{z} - \mathbf{E}\alpha^{(t)})). \tag{13}$$

$\lambda_t$ is a learnable thresholding vector for each vector entry in each layer. $\mathbf{W}$ and $\mathbf{E}$ perform the linear transformation.

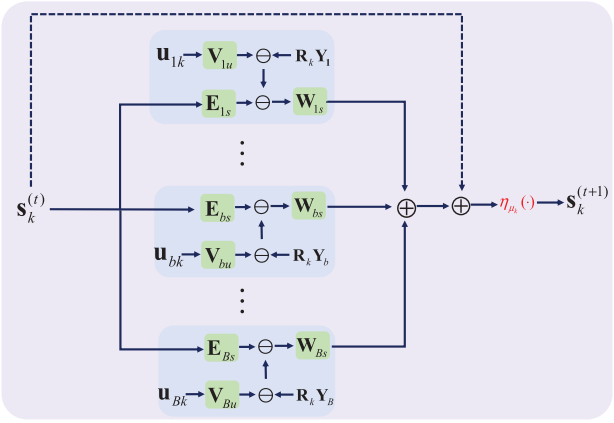



Fig. 2. SSR module corresponding to one iterative step for updating $\mathbf{s}_k$.

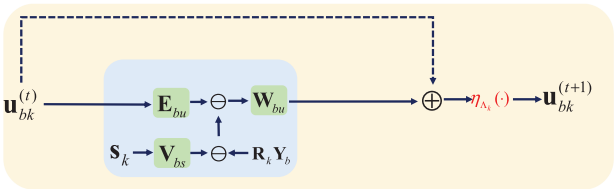



Fig. 3. USR module corresponding to one iterative step for updating $\mathbf{u}_{bk}$.

$\eta$ performs a nonlinear soft-thresholding operation to encourage sparsity. Each layer corresponds to one iteration in (7). Stacking the layer a fixed number of times yields the so-called learned ISTA (LISTA) network [55]. LISTA learns the sparse coding of the given input $\mathbf{z}$ in a supervised manner, and all the parameters are learned from the data rather than manually tuning. Once the network is trained, the sparse coding can be efficiently computed by the feedforward process.

Adopting a similar strategy, (9) and (11) can be accordingly converted into the SSR and USR modules.

*1) SSR Module:* SSR module aims to extract the common sparse structure among bands describing the global spectral correlation. By introducing $\mathbf{W}_{bs}$, $\mathbf{E}_{bs}$, and $\mathbf{V}_{bu}$, the update rule in (9) is converted into the SSR module whose definition is

$$\mathbf{s}_k^{(t+1)} = \eta_{\boldsymbol{\mu}_k}\left( \mathbf{s}_k^{(t)} + \sum_b \mathbf{W}_{bs}\left(\mathbf{R}_k\mathbf{Y}_b - \mathbf{V}_{bu}\mathbf{u}_{bk} - \mathbf{E}_{bs}\mathbf{s}_k^{(t)}\right)\right). \tag{14}$$

Fig. 2 illustrates the SSR module where $\{\mathbf{W}_{bs}, \mathbf{E}_{bs}, \mathbf{V}_{bu}, \boldsymbol{\mu}_k\}$ are the learnable parameters.

*2) USR Module:* Fig. 3 shows the USR module. The USR module is used to depict the unique sparse structure of individual bands. According to (11), for a specified band $b$, the USR module can be deduced as follows:

$$\mathbf{u}_{bk}^{(t+1)} = \eta_{\lambda_k}\left( \mathbf{u}_{bk}^{(t)} + \mathbf{W}_{bu}\left(\mathbf{R}_k\mathbf{Y}_b - \mathbf{V}_{bs}\mathbf{s}_k - \mathbf{E}_{bu}\mathbf{u}_{bk}^{(t)}\right)\right) \tag{15}$$

where $\{\mathbf{W}_{bu}, \mathbf{V}_{bs}, \lambda_k, \mathbf{E}_{bu}\}$ are the introduced learnable parameters.

*3) Image Reconstruction Module:* As presented in Fig. 4, the IR module reconstructs the denoised image from the

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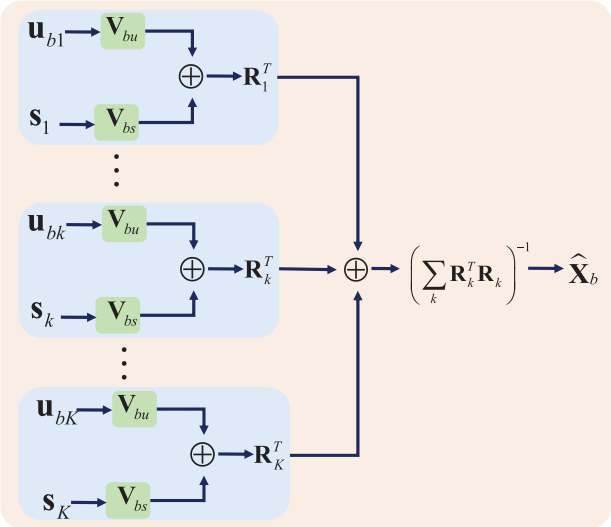



Fig. 4. IR module corresponding to reconstructing the clean HSI.

extracted $\{\mathbf{s}_k, \mathbf{u}_{bk}\}$. Based on (12), the module is given by

$$\widehat{\mathbf{X}}_b = \left(\sum_k \mathbf{R}_k^{\mathrm{T}} \mathbf{R}_k\right)^{-1} \sum_k \mathbf{R}_k^{\mathrm{T}} (\mathbf{V}_{bs}\mathbf{s}_k + \mathbf{V}_{bu}\mathbf{u}_{bk}) \tag{16}$$

where $\{\mathbf{V}_{bu}, \mathbf{V}_{bs}\}$ is the learnable parameter set.

### D. Implementation Details

In our implementation, we extract overlapping patches from the input HSIs for SR, with each one covering $9 \times 9$ pixels. The number of atoms in shared dictionaries ($m$) is 196, i.e., $\mathbf{A}_b \in \mathbb{R}^{81\times 196}$, and that of unshared dictionaries ($q$) is 64, i.e., $\mathbf{D}_b \in \mathbb{R}^{81\times 64}$. $\mathbf{A}_b$ and $\mathbf{D}_b$ are then used to initialize $\{\mathbf{W}_{bs}, \mathbf{V}_{bs}, \mathbf{E}_{bs}, \mathbf{W}_{bu}, \mathbf{V}_{bu}, \mathbf{E}_{bu}\}_{b=1\cdots B}$. $\{\boldsymbol{\mu}_t, \lambda_t\}_{t=1\cdots T}$ are initialized with 0.01. The number of unfolding layers $T$ is given as 5. We will exploit their impact on the denoising performance in our experiment section.

As our network is obtained by unfolding the optimization of the MTSR model, which embodies the physical properties of the HSIs, the network can be flexibly embedded with prior knowledge of HSIs. Specifically, we first run the "mexTrainDL" function in the SPAM[1] software on the training set by setting all the bands to share the same sparse coefficients, yielding the shared dictionary. The number of atoms in each training is given as desired. The other parameters are given as suggested in the software, i.e., sparsity promoting hyperparameter $\lambda$ is set to 0.15, and the total number of iterations is set to 1000. After that, we collect the dictionaries generated by each sample and cluster the dictionaries using the $K$-means algorithm whose cluster centers are considered as atoms in the dictionary. The dictionary is then used to initialize $\{\mathbf{W}_{bs}, \mathbf{V}_{bs}, \mathbf{E}_{bs}\}$. The unique dictionaries $\{\mathbf{W}_{bu}, \mathbf{V}_{bu}, and \mathbf{E}_{bu}\}$ are assigned as the basis of discrete cosine transform (DCT).

The loss function is set as the squared Euclidean distance between the estimated HSI and reference HSI. Given $N$ pair of training samples $\{\mathbf{Y}_i, \mathbf{X}_i\}_{i=1}^N$, the loss function is

$$\mathcal{L}(\Theta) = \frac{1}{N}\sum_{i=1}^{N} \| f(\Theta, \mathbf{Y}_i) - \mathbf{X}_i \|_F^2 \tag{17}$$

where $\Theta$ is the parameter set to be learned.

## IV. Experiments

In this section, we present the experimental setting, including the training datasets, compared methods, training details, and evaluation measures. After that, we extensively compare our methods with alternative methods, quantitatively and qualitatively. Moreover, a throughout network analysis is also given.

### A. Experimental Setting

*1) Training Setting:* The network is implemented in the PyTorch platform and trained with two NVIDIA GeForce RTX 3090 GPUs. The Adam optimizer is chosen to minimize the loss function with the batch size set as eight. Starting from $5 \times 10^{-4}$, the learning rate is cut to half with every ten epochs. It takes 100 epochs for network convergence. We trained multiple models, each designed to address specific noise scenarios.

*2) Compared Methods:* To comprehensively verify the denoising ability of our method, 11 competitive methods are used for comparison, including eight model-based methods and three DL-based methods. The model-based methods are BM4D [56], LRTDTV [57], LRMR [58], MTSNMF [40], LLRT [33], NGMeet [31], LRTF$L_0$ [22], and FastHyDe [28]. The DL-based methods are MAC-Net [47], MTSNN [17], and QRNN3D [24]. We try our best to tune the hyperparameters of the model-based methods to the optimal.

*3) Dataset Settings:* Two datasets are selected for evaluation, including ICVL and remote sensing HSIs. All the images were rescaled between 0 and 1 with min–max normalization where the minimum and maximum values were calculated across all the bands.

*a) ICVL dataset[2]:* The ICVL dataset is collected outdoor, with a Specim PS Kappa DX4 hyperspectral camera. The original image contains $1392 \times 1300$ pixels and 31 bands. Following [24], the training set includes 100 HSIs and the testing set includes 50 HSIs but with a different set to reduce the scene similarity between the training samples and testing ones. Specifically, we first cropped them to the size of $1024 \times 1024$. To enlarge the scale of the training set, we randomly crop subimages size of $64 \times 64 \times 31$ from the original image. We used three different settings for patch division: $1\times$, $0.5\times$, and $0.25\times$. In the $1\times$ setting, we extracted nonoverlapping patches from the original images. In the $0.5\times$ and $0.25\times$ settings, we rescaled the original images by a factor of 0.5 and 0.25, respectively, and then extracted patches with an overlap of 32 pixels. This methodology allowed us to obtain a total of 53 000 patches. Moreover, random flipping, cropping, and resizing are adopted to augment the training set. The main area with $512 \times 512$ pixels and 31 bands is used for testing.

[1]Link:http://thoth.inrialpes.fr/people/mairal/spams/

[2]http://icvl.cs.bgu.ac.il/hyperspectral/

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TABLE I
COMPARISON OF DIFFERENT METHODS ON 50 TESTING HSIS OF THE ICVL DATASET. THE TOP TWO VALUES ARE MARKED *Red* AND *Blue*

| | | | Model-based methods | | | | | | | | DL-based methods | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\sigma$ | Index | Noisy | BM 4D | MTS NMF | LRTD TV | LLRT | NG Meet | Fast HyDe | LRTF $L_0$ | LRMR | QRN N3D | MAC -Net | MTS NN | **MTS NN++** |
| **[0-15]** | PSNR | 33.18 | 44.39 | 45.38 | 41.54 | 45.74 | 39.63 | 48.08 | 43.41 | 41.50 | 47.84 | **48.21** | 47.18 | **48.86** |
| | SSIM | .6168 | .9683 | .9592 | .9617 | .9657 | .8612 | **.9917** | .9315 | .9356 | .9913 | **.9915** | .9887 | **.9917** |
| | SAM | .3368 | .0692 | .0845 | .0531 | .0832 | .2144 | .0404 | .0570 | .1289 | .0373 | .0387 | **.0371** | **.0346** |
| **[0-55]** | PSNR | 21.72 | 37.63 | 38.02 | 39.48 | 36.80 | 31.53 | 42.86 | 35.63 | 31.50 | **44.33** | 43.74 | 42.21 | **43.88** |
| | SSIM | .2339 | .9008 | .8586 | .9477 | .8285 | .6785 | **.9800** | .8125 | .6233 | **.9809** | .9768 | .9720 | .9794 |
| | SAM | .7012 | .1397 | .2340 | .0707 | .2316 | .4787 | .0630 | .1914 | .3583 | **.0553** | .0582 | .0614 | **.0528** |
| **[0-95]** | PSNR | 17.43 | 34.72 | 34.81 | 37.88 | 31.89 | 27.62 | 40.84 | 32.83 | 27.00 | 41.08 | **41.24** | 40.36 | **42.15** |
| | SSIM | .1540 | .8402 | .7997 | .9356 | .6885 | .5363 | **.9734** | .7482 | .4208 | .9541 | .9577 | .9610 | **.9720** |
| | SAM | .8893 | .1906 | .3266 | .0870 | .3444 | .6420 | .0771 | .3014 | .5142 | .0920 | .0841 | **.0758** | **.0665** |
| **Mixture** | PSNR | 13.21 | 23.36 | 27.55 | 34.21 | 18.23 | 23.61 | 27.58 | 30.93 | 23.10 | **38.43** | 28.44 | 36.66 | **38.90** |
| | SSIM | .0841 | .4275 | .6743 | .9023 | .1731 | .4448 | .7250 | .8378 | .3463 | .9302 | .7393 | **.9330** | **.9531** |
| | SAM | .9124 | .5476 | .5326 | .1396 | .6873 | .6252 | .4534 | .3613 | .5144 | .1154 | .4154 | **.1082** | **.0885** |

*b) Remote sensing dataset:* The remote sensing dataset includes synthetic Houston 2018 and real-world Gaofen-5 HSIs. The Houston 2018 HSI[3] describes the National Center for Airborne Laser Mapping (NCALM) at the University of Houston. There are 48 bands and 1202 × 4172 pixels. In our experiments, we chose the last 46 bands, which are relatively clean, and cropped a 512 × 512 cube from the center part of the HSI for testing and the rest for training. The same as the ICVL training set, we extracted patches with three settings. For the 1× images, we used patches with an overlap of 40 pixels. For the 0.5× and 0.25× images, we used patches with an overlap of 52 pixels. In total, we obtained 16 989 patches for the Houston 2018 HSI dataset. The real-world remote sensing dataset includes the Beijing Capital Airport HSI[4] and Pavia University HSI.[5] The Beijing Capital Airport HSI was imaged by Gaofen-5 satellite and contains 300 × 300 pixels and 155 bands. Pavia University HSI was acquired by the ROSIS sensor and includes 610 × 340 pixels and 103 bands.

*4) Evaluation Metrics:* All the methods are quantitatively evaluated under three standard metrics, i.e., peak signal-to-noise ratio (PSNR), structure similarity (SSIM), and spectral angle mapper (SAM). PSNR and SSIM are traditional indexes to quantify the reconstruction quality of lossy compression and perceived changes in structural information. SAM calculates the spectral angle between denoised and clean HSIs. The larger value of PSNR and SSIM and the smaller value of SAM indicate better denoising performance.

*5) Noise Settings:* To account for the varying noise variances across bands, we introduced spectrally non-i.i.d. noise to the clean HSIs to generate the noisy HSIs. Specifically, we selected noise levels in three different cases, with standard variances $\sigma$ ranging from [0–15], [0–55], and [0–95]. In addition, we designed a mixture noise case, where the standard variances $\sigma$ of Gaussian noise ranged from 0 to 95, with other noises as follows: 1) impulse noise on 1/3 of the bands, with intensities ranging from 10% to 70%; 2) strips on 5%–15% of columns on 1/3 of the bands; and 3) dead lines on 5%–15% of columns on 1/3 of the bands.

[3]https://hyperspectral.ee.uh.edu/

[4]http://hipag.whu.edu.cn/resourcesdownload.html

[5]https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

### B. Comparison With State-of-the-Art Methods

*1) Comparison on ICVL Dataset:* Table I summarizes a comparable result of all the methods on the ICVL testing dataset. It can be seen from the table that our method outperforms in different noise variances by achieving the best indexes in most cases. Specifically, among model-based methods, FastHyDe achieves the best denoising in the case of Gaussian noises, thanks to the simultaneous exploration of spatial–spectral correlation, spectral low-rankness, and spatial nonlocal self-similarity priors of HSIs. Attributing to the sparse term and global spectral low-rankness exploitation, LRTDTV and LRTF$L_0$ stand out in the mixture noises. It is interesting to see that MAC-Net shows a significant performance drop in the case of mixture noise. The main reason is that MAC-Net inherently assumes the noises follow Gaussian distribution to estimate the subspace. The mixture noises can hinder the accurate estimation of the space, leading to poor denoising performance. With only sparse and spatial–spectral priors of HSIs, our MTSNN++ greatly surpasses FastHyDe, LRTDTV, and LRTF$L_0$ by a large margin owing to data-driven training. Our method also provides superior performance than MTSNN. The main reason is that the advanced MTSR model can depict the shared and unshared structures of hyperspectral bands at the same time, helping recover both image contours and details. It is also interesting that our MTSNN++ provides less effective denoising than QRNN3D in the case of $\sigma \in$ [0–55]. The possible reason is that QRNN3D was gradually trained to denoise from low noise intensity to high noise intensity, which makes the learning process easier. However, our MTSNN++ still outperforms QRNN3D in other cases.

We also evaluate the visual quality of all the compared methods in Figs. 5 and 6. A representative HSI *lehavim_0910-1605* is used for assessment. From Fig. 5, MTSNMF and

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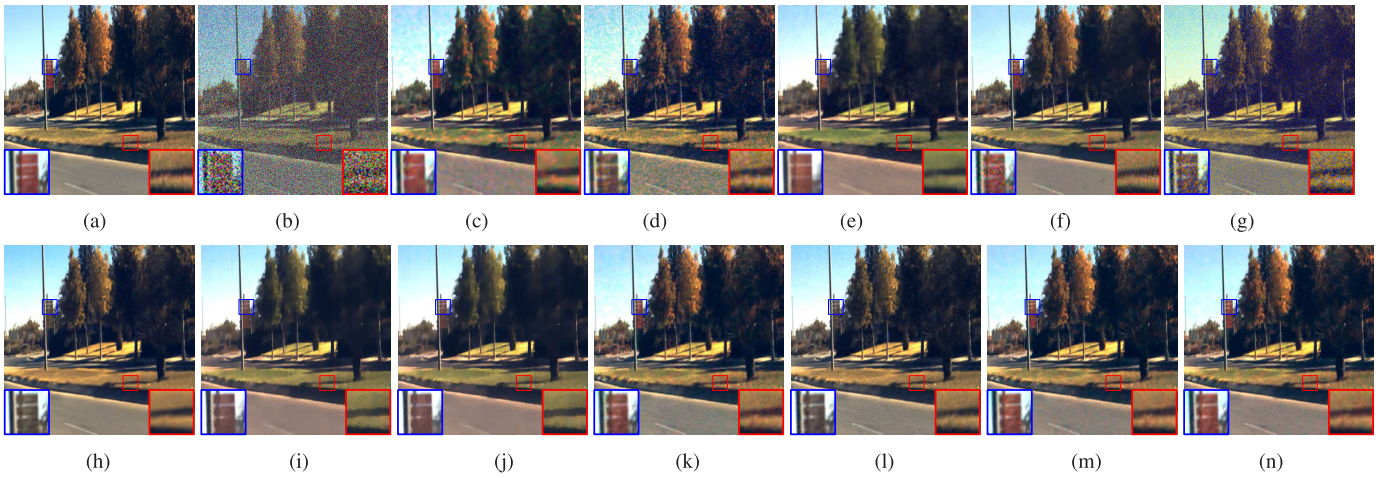



Fig. 5. Denoising results on the *lehavim_0910-1605* image in the ICVL dataset with the noise variance in [0–95]. The false-color images were generated by combining bands 31, 21, and 9 **(zoom in to see details)**. (a) Clean. (b) Noisy. (c) BM4D. (d) MTSNMF. (e) LRTDTV. (f) LLRT. (g) NGMeet. (h) FastHyDe. (i) LRTF$L_0$. (j) LRMR. (k) QRNN3D. (l) MAC-Net. (m) MTSNN. (n) MTSNN++.

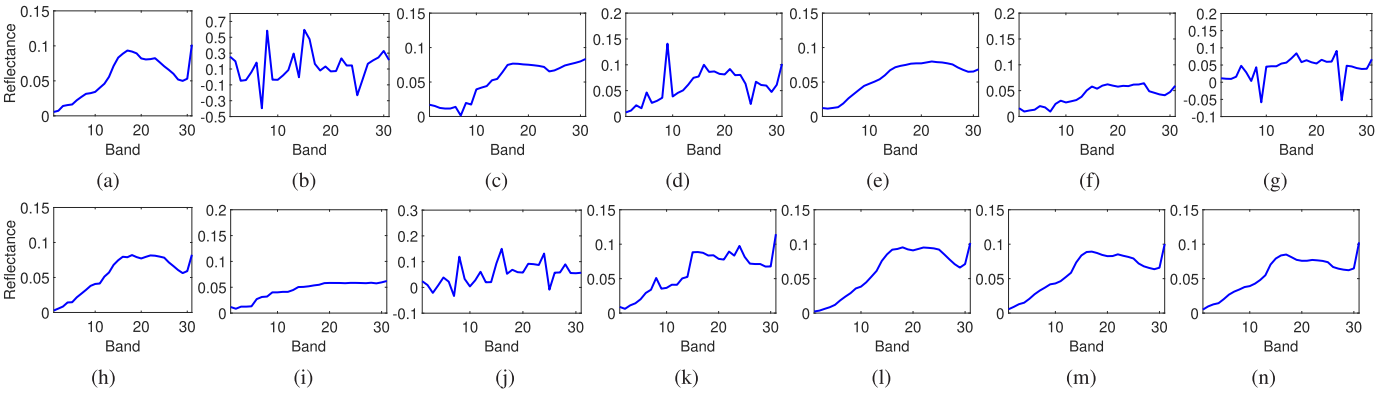



Fig. 6. Spectrum of pixel (430, 200) in *lehavim_0910-1605* image with the noise variance in [0–95] **(zoom in to see details)**. (a) Clean. (b) Noisy. (c) BM4D. (d) MTSNMF. (e) LRTDTV. (f) LLRT. (g) NGMeet. (h) FastHyDe. (i) LRTF$L_0$. (j) LRMR. (k) QRNN3D. (l) MAC-Net. (m) MTSNN. (n) MTSNN++.

NGMeet cannot fully remove all the noise because of the difficulties of hyperparameter selection. LLRT and BM4D produce obvious artifacts. LRTF$L_0$ loses many important details. Compared with FastHyDe and MAC-Net, our MTSNN++ produces higher fidelity with the clean HSI by recovering more details and sharing a similar color distribution with the reference one. In general, the clean spectrum of each pixel is continuous between bands, as can also be observed in Fig. 6(a). Because of the noise contamination, the continuous property can be destroyed; see Fig. 6(b). Limited by the poor denoising ability, MTSNMF, and NGMeet cannot recover the clean spectrum and result in significant peaks. Thanks to the advanced physical model and data-driven training, our MTSNN++ is more capable of recovering the clean spectrum by a good match with the clean one. To summarize, this experiment evidently shows the high denoising capacity of the proposed MTSNN++, quantitatively and qualitatively.

*2) Comparison on Houston 2018 Dataset:* In this experiment, we fine-tuned QRNN3D and MAC-Net using the model previously trained on the ICVL dataset. However, due to the Houston dataset containing 46 bands, which is incompatible with the ICVL dataset with 31 bands, we directly trained MTSNN and MTSNN++ on Houston HSIs. The denoising

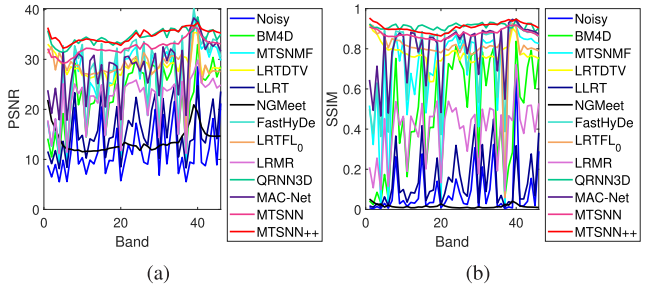



Fig. 7. Bandwise comparison on the Houston 2018 HSI in the mixture noise case. (a) PSNR. (b) SSIM.

performance of all the methods is presented in Table II. Overall, our MTSNN++ achieves the most promising denoising results across all the scenarios. Moreover, compared with the baseline MTSNN, our MTSNN++ consistently demonstrates higher performance in all the cases, which can be attributed to its improved physical modeling. We further plot the bandwise PSNR and SSIM with respect to the mixture noise case in Fig. 7. In general, MTSNN++ and QRNN3D achieve the best bandwise indices. Compared with QRNN3D, our MTSNN++ demonstrates competitive marginal PSNR but lower marginal SSIM. This difference may be attributed to

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TABLE II

COMPARISON OF DIFFERENT METHODS ON THE HOUSTON 2018 HSI. THE TOP TWO VALUES ARE MARKED *Red* AND *Blue*

| | | | **Model-based methods** | | | | | | | | **DL-based methods** | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\sigma$ | Index | Noisy | BM 4D | MTS NMF | LRTD TV | LLRT | NG Meet | Fast HyDe | LRTF $L_0$ | LRMR | QRN N3D | MAC-Net | **MT SNN** | **MT SNN++** |
| **[0-15]** | PSNR | 32.47 | 40.70 | 42.83 | 31.10 | 40.65 | 38.31 | 45.87 | 42.74 | 39.90 | 44.79 | **48.13** | 44.85 | **48.11** |
| | SSIM | .6738 | .9667 | .9775 | .8286 | .9468 | .8906 | **.9927** | .9842 | .9601 | .9878 | **.9945** | .9862 | **.9927** |
| | SAM | .2562 | .0550 | .0357 | .0955 | .0712 | .1279 | .0355 | .0420 | .0612 | .0315 | **.0269** | .0285 | **.0232** |
| **[0-55]** | PSNR | 24.66 | 35.04 | 35.94 | 30.86 | 30.15 | 29.86 | **43.99** | 36.99 | 30.93 | 41.61 | 43.67 | 39.58 | **44.98** |
| | SSIM | .3874 | .9002 | .8919 | .8219 | .6390 | .6918 | **.9888** | .9386 | .7624 | .9747 | **.9878** | .9631 | .9862 |
| | SAM | .6484 | .1014 | .1387 | .1006 | .3295 | .3707 | **.0391** | .1124 | .1688 | .0436 | .0406 | .0415 | **.0292** |
| **[0-95]** | PSNR | 16.85 | 31.23 | 32.73 | 30.11 | 22.93 | 26.06 | **39.32** | 33.86 | 26.43 | 37.46 | **39.32** | 36.62 | **41.07** |
| | SSIM | .2072 | .7766 | .8160 | .8077 | .3724 | .5514 | **.9729** | .8730 | .5662 | .9304 | .9515 | .9708 | **.9719** |
| | SAM | .9201 | .1961 | .2160 | .1242 | .5802 | .5400 | .0715 | .1890 | .3017 | .0786 | .0662 | **.0442** | **.0434** |
| **Mixture** | PSNR | 11.72 | 22.76 | 25.86 | 28.60 | 15.58 | 22.36 | 27.07 | 28.75 | 21.84 | **34.42** | 28.38 | 32.24 | **34.71** |
| | SSIM | .0843 | .4762 | .6933 | .7784 | .1386 | .5169 | .7757 | .8038 | .3914 | **.9121** | .8028 | .8729 | **.9081** |
| | SAM | .9778 | .5168 | .4977 | .1433 | .7652 | .5728 | .4518 | .2221 | .4857 | **.0928** | .2343 | .1119 | **.0817** |

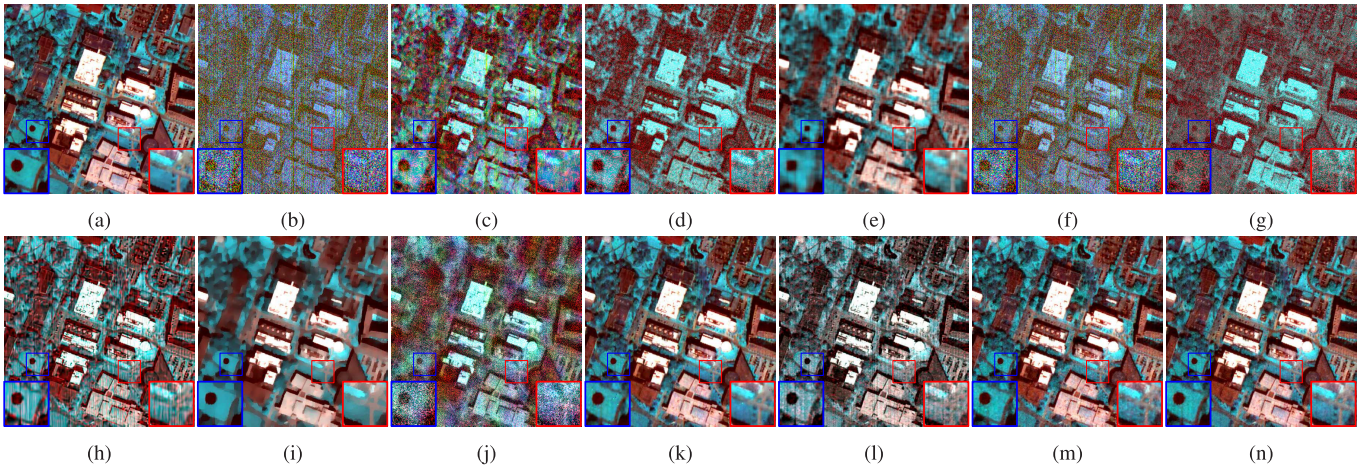


Fig. 8. Denoising results on the Houston 2018 HSI in the mixture noise case. The false-color images were generated by combining bands 4, 32, and 46 **(zoom in to see details)**. (a) Clean. (b) Noisy. (c) BM4D. (d) MTSNMF. (e) LRTDTV. (f) LLRT. (g) NGMeet. (h) FastHyDe. (i) LRTF$L_0$. (j) LRMR. (k) QRNN3D. (l) MAC-Net. (m) MTSNN. (n) MTSNN++.

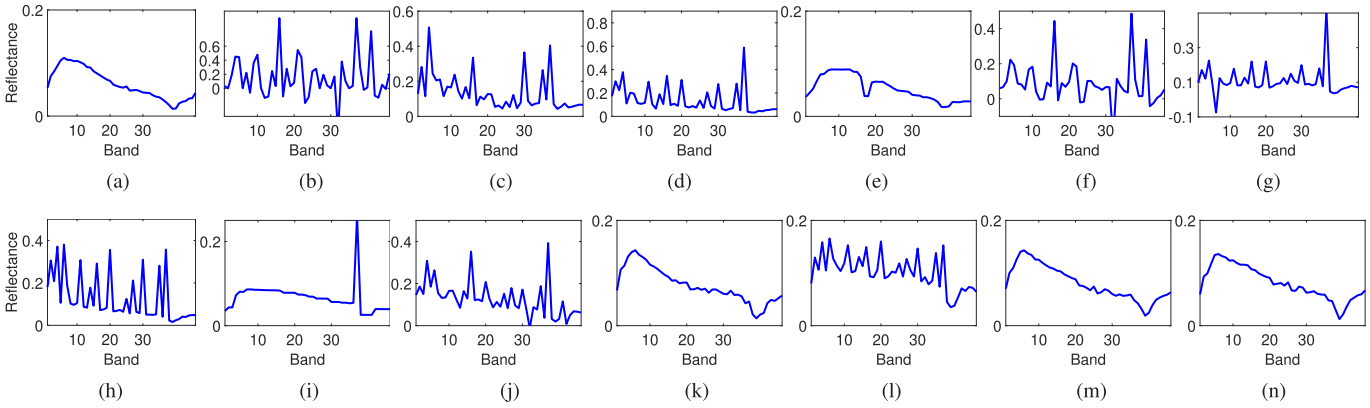



Fig. 9. Spectrum of pixel (208, 245) in the Houston 2018 HSI in the mixture noise case before and after denoising **(zoom in to see details)**. (a) Clean. (b) Noisy. (c) BM4D. (d) MTSNMF. (e) LRTDTV. (f) LLRT. (g) NGMeet. (h) FastHyDe. (i) LRTF$L_0$. (j) LRMR. (k) QRNN3D. (l) MAC-Net. (m) MTSNN. (n) MTSNN++.

the fact that QRNN3D fine-tunes the model trained on the non-i.i.d. Gaussian case for the mixture noise scenario, while our MTSNN++ directly learns the denoiser for mixture noise scenario, which is a more challenging task.

In Fig. 8, we showcase the denoising results for the case of mixture noise. It is a challenging task to remove all the mixture noises, and thus they remain in most methods. LRTDTV and LRTF$L_0$ produce oversmoothed images due to the challenge

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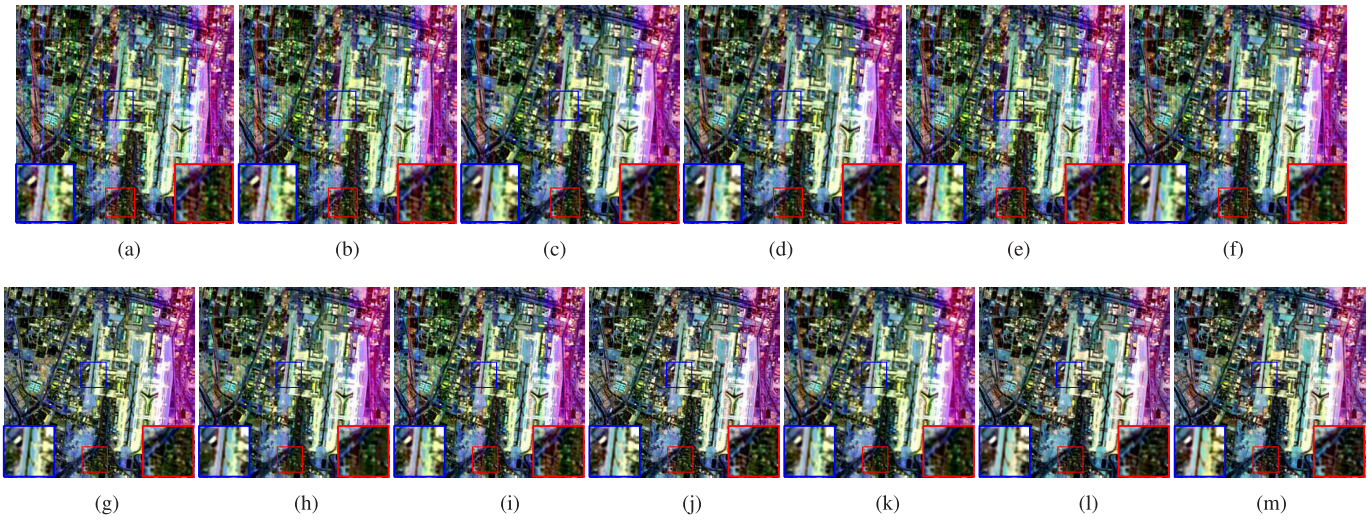



Fig. 10. Denoising results on the real-world Beijing Capital Airport HSI. The false-color images were generated by combining bands 152, 88, and 42. (a) Noisy. (b) BM4D. (c) MTSNMF. (d) LRTDTV. (e) LLRT. (f) NGMeet. (g) FastHyDe. (h) LRTF$L_0$. (i) LRMR. (j) QRNN3D. (k) MAC-Net. (l) MTSNN. (m) MTSNN++.

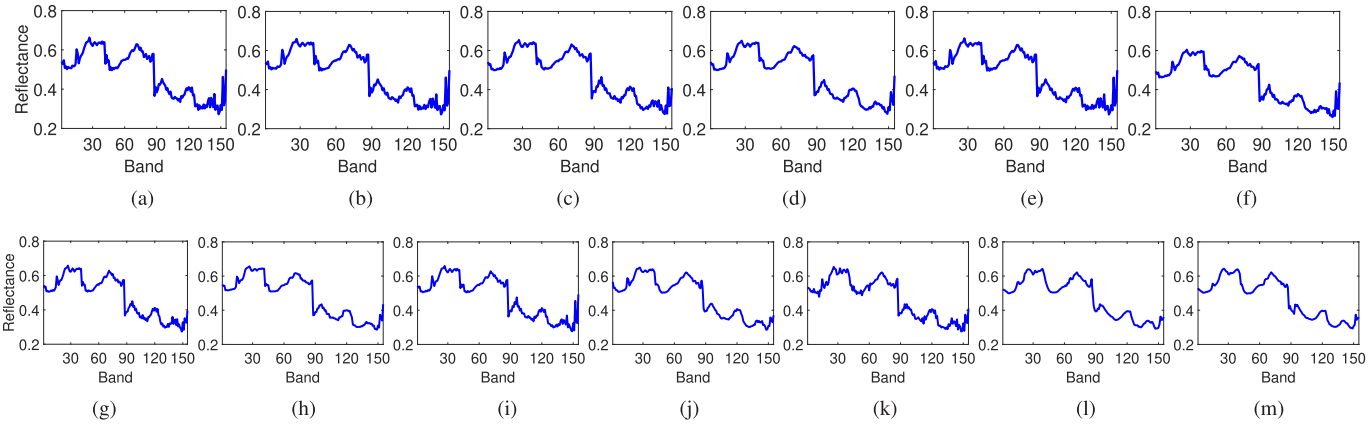



Fig. 11. Spectrum of pixel (258, 192) in the Beijing Capital Airport HSI before and after denoising **(zoom in to see details)**. (a) Noisy. (b) BM4D. (c) MTSNMF. (d) LRTDTV. (e) LLRT. (f) NGMeet. (g) FastHyDe. (h) LRTF$L_0$. (i) LRMR. (j) QRNN3D. (k) MAC-Net. (l) MTSNN. (m) MTSNN++.

of finding the right balance between noise reduction and preservation of details when setting the hyperparameters. Our MTSNN++ reduces most of the noise while preserving complex textures. Fig. 9 displays the spectral reflectance of pixel (341, 166). The performance of learning-based methods is superior to that of model-based methods. Among them, MTSNN++ is more capable of recovering the spectrum from the disordered observation.

*3) Comparison on Real-World Beijing Capital Airport HSI:* We applied the trained models from the close-range ICVL dataset directly to test the Beijing Capital Airport HSIs, as these models were trained with more training samples. We split the HSIs into multiple nonoverlapping subband HSIs, with each subband HSI containing 31 bands to test MTSNN and MTSNN++. As there are no clean counterparts for real-world remote sensing HSIs, we compared the denoising results qualitatively with respect to band images and the recovered spectrum.

As shown in Fig. 10, the image is contaminated with obvious noise. Residual noise still exists in the results of BM4D and LLRT. The main reason is that they are specifically developed for Gaussian noise and cannot handle complicated mixture noise. The LRTF$L_0$ tends to lose many details due to the oversmoothing effect caused by the $L_0$-norm-based total variation which focuses on smoothing the image to preserve the important edges. MAC-Net also results in the loss of image details, particularly in the airplane target area. This could be attributed to the ignorance of the band-specific properties of HSIs, which are crucial for preserving fine details and accurately representing the spectral characteristics of the target. Compared with other learning-based methods, our MTSNN++ produces visually more satisfying results with fine details and important edges. Overall, the performance of all the methods is comparable. We further present the recovered spectrum of pixel (258, 192) for all the methods in Fig. 11. Compared with other methods, MTSNN and MTSNN++ achieve more continuous spectrum. The good performance of our MTSNN++ mainly attributes to two factors. On one hand, our MTSR model simultaneously captures the shared common and unshared unique sparse structure of HSIs, facilitating the removal of noise while preserving edges and textures. On the other hand, the unfolded network inherits the

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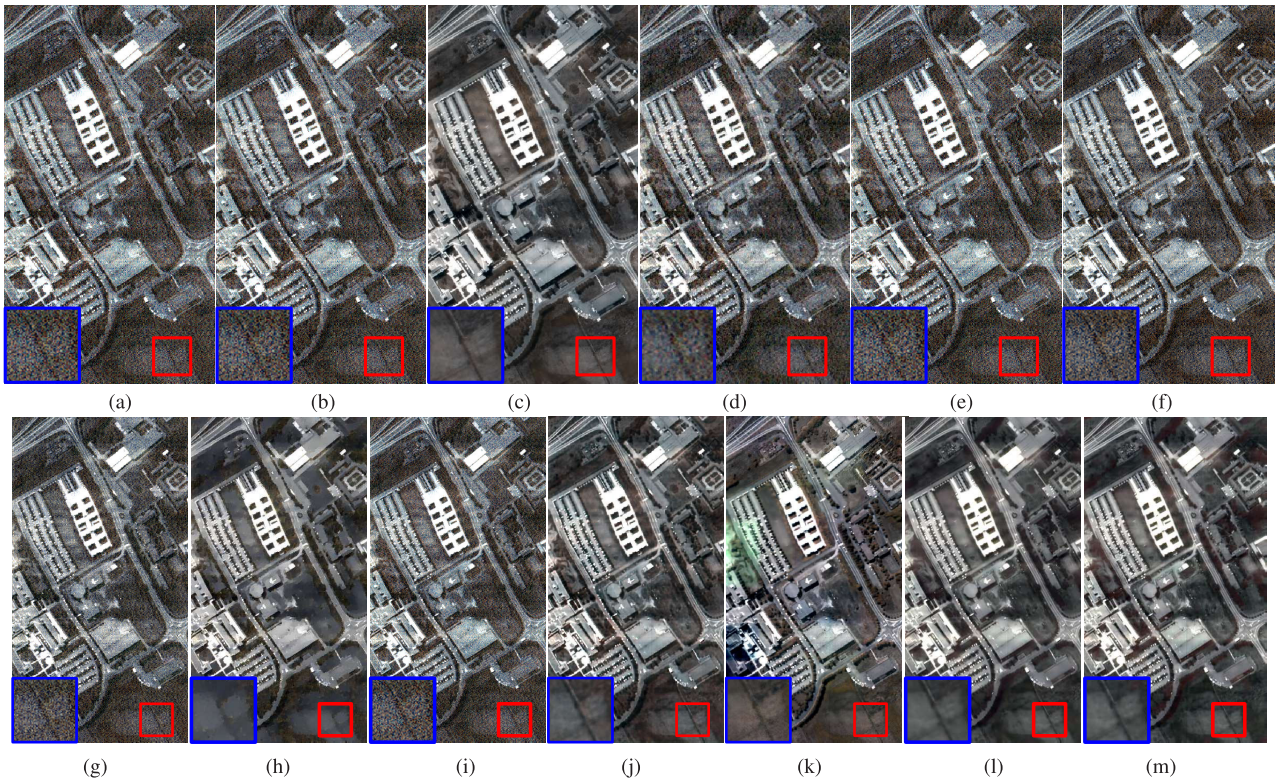


Fig. 12. Denoising results on the **real-world Pavia University HSI**. The false-color images were generated by combining bands 1–3 **(zoom in to see details)**. (a) Noisy. (b) BM4D. (c) MTSNMF. (d) LRTDTV. (e) LLRT. (f) NGMeet. (g) FastHyDe. (h) LRTF$L_0$. (i) LRMR. (j) QRNN3D. (k) MAC-Net. (l) MTSNN. (m) MTSNN++.

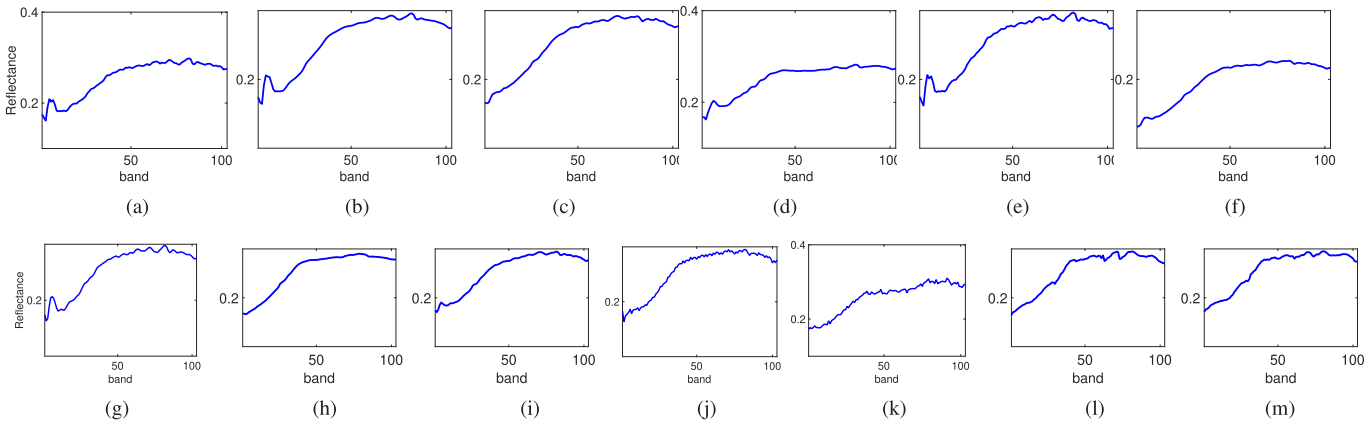



Fig. 13. Spectrum of pixel (160, 416) in the **real-world Pavia University HSI (zoom in to see details)**. (a) Noisy. (b) BM4D. (c) MTSNMF. (d) LRTDTV. (e) LLRT. (f) NGMeet. (g) FastHyDe. (h) LRTF$L_0$. (i) LRMR. (j) QRNN3D. (k) MAC-Net. (l) MTSNN. (m) MTSNN++.

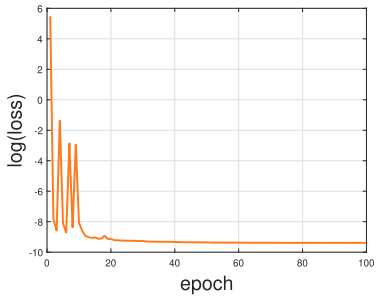



Fig. 14. Loss changes during training on the ICVL dataset.

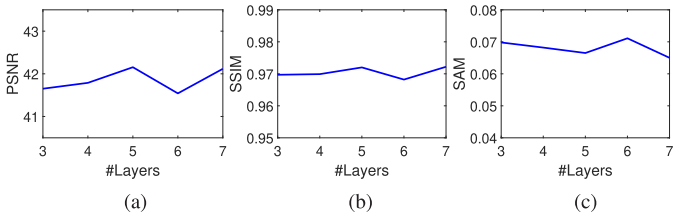
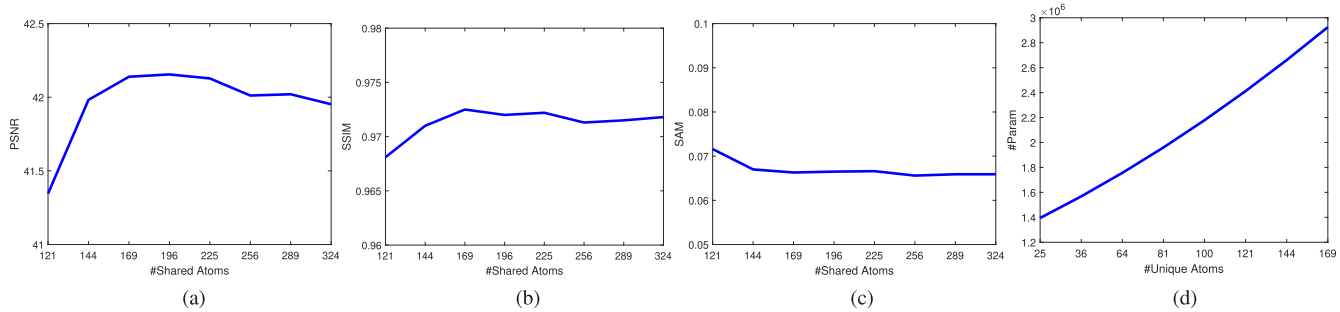



Fig. 15. Impact of the number of unfolding layers on the denoising performance. (a) PSNR. (b) SSIM. (c) SAM.

advantages of the physical model and has higher representation ability, thanks to data-driven training.

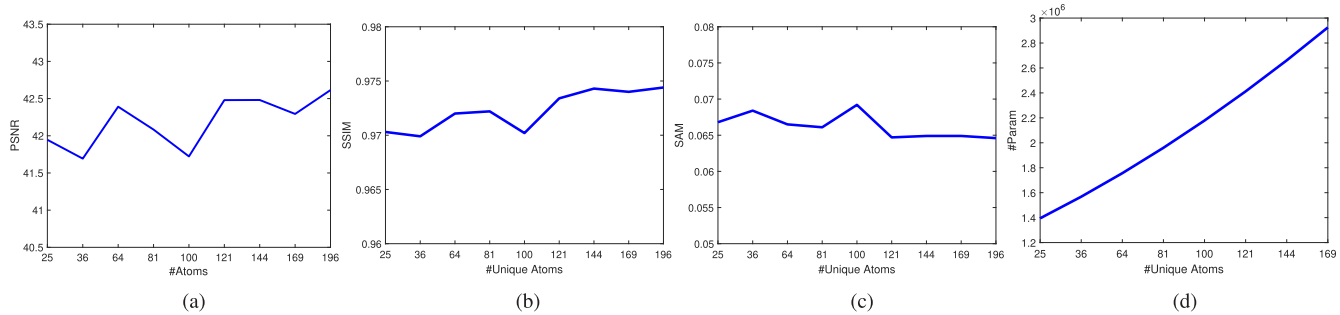
*4) Comparison on Real-World Pavia University HSI:* The first three bands of Pavia University HSI are contaminated with noise. The denoising results of bands 1–3 are shown

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(a) (b) (c) (d)

Fig. 16. Impact of the number of atoms in shared dictionaries for each band on the denoising performance. (a) PSNR. (b) SSIM. (c) SAM. (d) #Param.




(a) (b) (c) (d)

Fig. 17. Impact of the number of atoms in unique dictionaries for each band on the denoising performance. (a) PSNR. (b) SSIM. (c) SAM. (d) #Param.

in Fig. 12. MTSNMF, QRNN3D, MAC-Net, MTSNN, and MTSNN++ are more capable of removing noises because they can better exploit the spatial–spectral correlation of HSIs. Comparatively, MTSNN++ can successfully recover sharper borders. This attributes to the shared sparse structure modeling among bands which facilitates recovering the predominant edges. However, MTSNN++ also misses some details on the tree target than MAC-Net. It is likely that MAC-Net explores the spatial nonlocal self-similarity of HSIs, which benefits preserving image details. Fig. 13 demonstrates the reconstructed spectrum at pixel (160, 416). Our MTSNN++ can eliminate the noise in the first few bands by presenting continuous curves while keeping the curves in the remaining bands as the original curves.

### C. Network Analysis

In this section, we investigate the key components of the network, including the dictionary settings and the number of layers on the denoising performance. All the experiments are conducted with $\sigma \in [0–95]$.

*1) Loss Changes During Network Training:* We plot the average training loss of MTSNN++ in Fig. 14. As can be seen, the network fluctuates in the first few training epochs because of stochastic gradient descent and gradually reaches convergence.

*2) Impact of the Number of Layers:* We further change the number of unfolding layers ($T$) from three to seven with an interval of 1 to further explore the influence of the number of unfolding layers on the denoising performance. As shown in Fig. 15, the denoising effectiveness becomes higher when $T$ changes from 3 to 5. After peaking at 5, the denoising effectiveness drops. The possible reason is that the network is obtained by unfolding the recurrent neural network. As the number of layers increases, it becomes more challenging to train a robust network. For this reason, the number of layers is suggested to be set as five.

*3) Impact of Dictionary Settings:* We first set the number of atoms in unique dictionaries as 64 and change the number in shared dictionaries from 121 to 324 to show the denoising performance changes. From Fig. 16, the best denoising performance is achieved when the number is set as 196. When the number reaches 324, the number of parameters is almost 1.5 times higher than when it is set to 196, but the performance gain is insignificant. Under this observation, the number of atoms in shared dictionaries for each band is given as 196.

Second, we examine the number of atoms in unique dictionaries on the denoising effect in Fig. 17. Overall, the denoising performance improves with the number of atoms growing. More atoms also lead to more parameters being learned and a higher computational load. When the number of atoms changes from 196 to 64, the PSNR value only decreases by about 0.34 dB, but the number of parameters decreases by about 1 000 000. For this reason, we set the number of unique dictionaries for each band as 64, balancing the denoising performance and computational complexity.

Third, to gain more insight into the network, we report the learned parameters of some representative bands in Fig. 18. By observing the figure, there are significant differences between the initialized parameters and the learned parameters, except for $\mathbf{E}_{bu}$. This indicates that the network aims to learn more appropriate parameters that can better adapt to the data.

From the figure, $\mathbf{W}$, $\mathbf{V}$, and $\mathbf{E}$ are different in both shared and unshared dictionaries, meaning that their roles are different in the network. This is also likely the reason why a fixed

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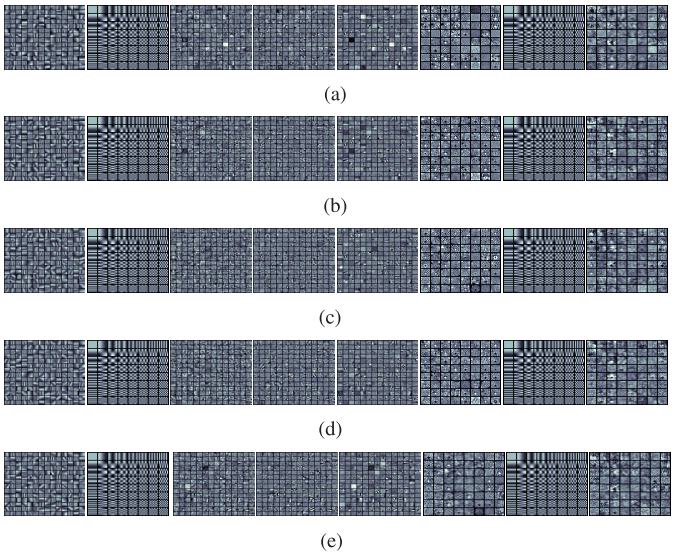


Fig. 18. Demonstration of learned parameters. From left to right are initialized shared dictionary, unique dictionary, $\mathbf{W}_{bs}$, $\mathbf{E}_{bs}$, $\mathbf{V}_{bs}$, $\mathbf{W}_{bu}$, $\mathbf{E}_{bu}$, and $\mathbf{V}_{bu}$, respectively **(zoom in to see details)**. (a) Band 1. (b) Band 7. (c) Band 14. (d) Band 21. (e) Band 28.

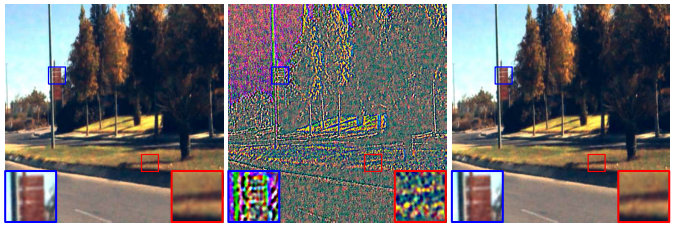


Fig. 19. Reconstructed HSI from different modules. (a) SSR (37.88 dB). (b) USR (11.79 dB). (c) SSR+USR (38.04 dB).

number of layers can fulfill a large number of iterations in traditional optimization. On the other hand, we can see that the shared dictionaries mainly contain sharp image edges indicating coarse image information. Differently, the unique structures mainly imply fine details. The main reason is the physical model enforces all the bands to share the same coefficients to depict the global correlation and adopt different coefficients to depict the unique properties of individual bands. To fulfill this goal, the shared dictionaries should include the global structures related to image edges, and unshared dictionaries should contain fine structures related to image details. To gain further intuition, we present the image reconstructed from the SSR and USR modules in Fig. 19. The image obtained from the SSR module contains the main image contents, while that obtained from the USR module is related to image details.

We further change the initialization strategy of the shared dictionary from the one obtained from the SPAM software to the analytical DCT basis to show the effect of network initialization. As can be seen from Table III, the use of the dictionaries trained by the SPAM software can boost denoising performance. The main reason is that the learned dictionaries are more adaptive than the analytical DCT.

*4) Ablation Study on SSR/USR Modules:* When the USR module is removed from the network, MTSNN++ becomes MTSNN. Therefore, we directly use the denoising performance of MTSNN in this case. As the case of removing the SSR module, MTSNN++ becomes a network that performs denoising in a band-by-band manner. We set the number of atoms as 256 to train such a network, considering that such setting gives the best denoising performance.

As observed from Table IV, removal of the SSR module results in global spectral correlation not being used effectively. Removing the USR module causes the band-specific property to be ignored. Simultaneous consideration of the global and local spectral correlation produces the best denoising effect.

TABLE III
IMPACT OF THE NETWORK INITIALIZATION

| Initialization | PSNR | SSIM | SAM |
|---|---|---|---|
| SPAM | **42.39** | **.9788** | **.0252** |
| DCT | 42.21 | .9781 | .0259 |

TABLE IV
ABLATION STUDY ON SSR AND USR MODULES

| SSR | USR | PSNR | SSIM | SAM |
|---|---|---|---|---|
| ✓ | ✗ | 40.36 | .9610 | .0758 |
| ✗ | ✓ | 34.72 | .9003 | .0794 |
| ✓ | ✓ | **42.15** | **.9720** | **.0665** |

## V. CONCLUSION

This article addresses the HSI denoising problem with the MTSR model-inspired neural network. We have shown that the network is able to learn the shared sparse coefficients to depict common structures related to image edges and unshared sparse coefficients to describe the fine image details. Experiments on synthetic data and real HSIs have demonstrated that our method can outperform both the model-based and DL-based methods. We will explore the network in depth by incorporating other priors such as nonlocal self-similarity and low-rankness to enhance HSI denoising further.

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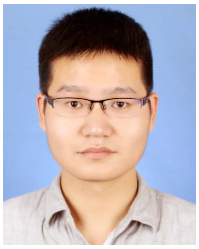
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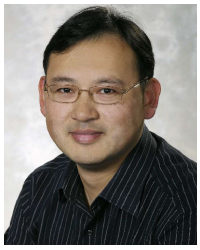
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